

R TO PYTHON ROSETTA

The map from what you know to what you are typing.

IN R	IN PYTHON	WHAT IT IS	IN R	IN PYTHON	WHAT IT IS
<code>x <- 3</code>	<code>x = 3</code>	assign	<code>table(x)</code>	<code>x.value_counts()</code>	counts
<code>c(1, 2, 3)</code>	<code>[1, 2, 3]</code>	a list	<code>df[df\$x > 1,]</code>	<code>df[df['x'] > 1]</code>	filter rows
<code>length(x)</code>	<code>len(x)</code>	how many	<code>function(x) ...</code>	<code>def f(x): ...</code>	a function
<code>x[1]</code>	<code>x[0]</code>	first item, Python counts from 0	<code>sapply(x, f)</code>	<code>[f(i) for i in x]</code>	a comprehension
<code>data.frame</code>	<code>DataFrame</code>	a table, from pandas	<code>x %>% f()</code>	<code>x.f()</code>	method chaining
<code>df\$col</code>	<code>df['col']</code>	one column	<code>library(dplyr)</code>	<code>import pandas as pd</code>	load a package
<code>nrow(df)</code>	<code>len(df)</code>	row count	<code>paste0('a', 'b')</code>	<code>f'{a}{b}'</code>	join strings
<code>head(df)</code>	<code>df.head()</code>	first rows	<code>is.na(x)</code>	<code>x.isna()</code>	missing test
<code>str(df)</code>	<code>df.info()</code>	structure	<code>NA</code>	<code>None</code> or <code>NaN</code>	missing
<code>summary(df)</code>	<code>df.describe()</code>	summary stats	<code>TRUE / FALSE</code>	<code>True / False</code>	booleans

Four things that will catch you

- Python counts from 0. The first item is `x[0]`, and `x[0:2]` gives you two items, stopping before index 2.
- Indentation is syntax. The block under a `for` or `if` is whatever is indented beneath it, so a stray space is an error.
- There is no `<-`. Assignment is `=`, and `==` is the test for equality.
- A dictionary is the object R has no clean equivalent for. Write it `{'left': 0, 'right': 1}` and read it with `d['left']`. Label maps, crosswalks and config all live in one.